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Andrei Beliaev

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Experience

Epivara, Inc., BioTech — Champaign, IL
AI/ML Software Engineer — *May 2025 – Present*

- Built and pretrained DINOv3 ViT + CNN-adapter model capturing tissue-scale structure down to sub-nuclear detail: +18.5% mSA and +4.6% F1 over a state-of-the-art baseline on Leydig-cell histology; underpinned a toxicology study.
- Engineered a 16× larger field of view at 2× lower inference cost and 3.7× faster training over a state-of-the-art baseline, supplying the anatomical context needed to reject biologically implausible classifications across ∼3/4 of surveyed tissue endpoints.
- Built an animal perception system with SAM 3.1 as the teacher, a DEIMv2-X student detector, DINOv3 re-identification (95.1% accuracy, 92.3% macro F1), and identity tracking through occlusion and re-entry.
- Developing 3D pose and behavior measurement for near-identical unmarked animals across synchronized machine-vision cameras; benchmarking multi-animal pose models under contact and occlusion.
- Shipped production agentic systems: an LLM tool-calling analysis platform that explores schemas, writes SQL, and runs sandboxed Python, plus a human-in-the-loop ingestion agent that reduced experimental-data insertion from hours to minutes.

Tinkoff, FinTech — Moscow, Russia
Data Scientist, Compliance — *Feb 2022 – Oct 2022*

- Contributed to feature development for a new Anti-Money Laundering model. The model increased fully automated case resolution from 45% to 60% and cut annual costs by $1.7M.
- Developed SQL/Python/Spark pipelines and dashboards that improved fraud and compliance analysis, cutting 1,000 hours of manual work annually.

Data Analyst, Insurance — *Mar 2021 – Feb 2022*

- Engineered a real-time quality measurement framework that merged backend logs, frontend events, and database tables, exposing live SRE metrics in Grafana and raising product quality (aggregated metric) by 18% while saving $60K monthly.
- Delivered ML-ready datasets for churn prediction, ran A/B tests, and worked with product and engineering teams to diagnose production issues.

Projects

HistoForge: Open-Source AI Platform for Digital Pathology (Launching Sep 2026) — [Link]

- Building an end-to-end platform (QuPath Java plugin + Modal GPU inference runtime + Hugging Face model hub + pip package) that brings foundation-model tissue analysis to pathologists and researchers.
- Pretrained a large field-of-view DINOv3 + CNN-adapter model on 16 public cell-segmentation datasets; shipping 25 nuclei- and structure-level modules trained on public and proprietary data.
- Packaging training and inference so labs can fine-tune, run, and share their own models and pipelines, targeting reproducible quantitative pathology for oncology and toxicology research.

Improving System-2 Thinking in Energy-Based Transformers — [Link]

- Ran MCMC sampling ablations in Energy-Based Transformers, comparing perplexity across pretraining configurations.

Image Immunization Against Diffusion-Based Editing

- Implemented image-immunization methods from CVPR 2024 and ICLR 2026, using timestep-universal gradients and learned perturbation networks for Stable Diffusion.

Entropy-aware sampling in vLLM — [Link]

- Implemented entropy-aware token sampling in vLLM with GPU-batched lookahead to control diversity by penalizing entropy-reducing tokens in speculative decoding.

Education

University of Illinois Urbana–Champaign — Champaign, IL
Master of Computer Science | Graduate Research Assistant — May 2026

Bauman Moscow State Technical University — Moscow, Russia
Specialist in Autonomous Informational and Control Systems — Jun 2022

Skills

Languages: Python (FastAPI, SQLAlchemy, asyncio), SQL, TypeScript (React, Zustand, shadcn/ui), Java, C++, MATLAB
AI/ML: PyTorch, LangGraph, vLLM, Scikit-learn, W&B, tool and agent orchestration, HITL, RAG
Infra/Data: AWS (EC2, S3, IAM, ECS, VPC), Modal, Docker, PySpark, PostgreSQL, HPC (SLURM)